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BERTRetrax: A Transformer-Augmented Pipeline for Retraction Detection in Scholarly Articles

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Abstract. The rising prevalence of retracted scholarly articles has elicited concerns about research integrity and the distribution of dubious scientific findings. This study introduces **BERTRetrax**, a transformer-enhanced pipeline aimed at automated identification of retracted research papers. The framework integrates classical machine learning algorithms—such as Logistic Regression, Random Forest, K-Nearest Neighbors, and AdaBoost with advanced transformer-based models including BERT, RoBERTa, ALBERT, DistilBERT, Electra, SciBERT, MobileBERT, and BioBERT. A substantial, empirical dataset comprising over 50,000 entries from Retraction Watch was utilized to assess the efficacy of each model. Comprehensive preprocessing and hyperparameter tuning were implemented, and performance was evaluated employing accuracy, precision, recall, F1-score, and confusion matrices. **BioBERT attained the highest classification accuracy of 98.23%** amongst all models, surpassing both general-purpose and domain-adapted transformer models. The findings highlight the efficacy of transformer models, especially those pre-trained on biomedical literature, in detecting retracted content. BERTRetrax exhibits exceptional performance and versatility, providing a resilient and scalable solution to assist publishers, reviewers, and research institutions in maintaining academic integrity and responsibility.

Keywords: Retraction Detection · Transformer Models · BioBERT · Text Classification · Scientific Integrity

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1 Introduction

The exponential spike in scientific literature has coincided with a surge in retracted publications, jeopardizing the integrity of academic research and potentially facilitating the spread of misinformation, particularly in critical domains such as medicine, biology, and social sciences [1]. Retractions frequently arise from diverse problems such as methodological deficiencies, data fabrication, plagiarism, and ethical violations, underscoring the pressing necessity for automated systems to identify such articles prior to the dissemination of their conclusions [2].

The manual identification and verification of retracted material are arduous, time-consuming, and susceptible to errors. Databases like **Retraction Watch** and **PubMed** provide curated compilations of retracted publications; nevertheless, these tools depend on post-publication notifications and are unable to proactively detect dubious research [3, 4]. As a consequence, researchers have employed **Natural Language Processing (NLP)** and **Machine Learning (ML)** methodologies to classify and identify potentially retracted publications utilizing metadata, textual content, and citation patterns [5].

Recent advancements in deep learning and **transformer-based architectures**, notably the **Bidirectional Encoder Representations from Transformers (BERT)** and its specialized variants such as **SciBERT**, **BioBERT** etc., have transformed text classification tasks by providing contextual comprehension that surpasses that of conventional machine learning models [6–8]. These models have demonstrated potential in several biological and academic NLP applications, such as paper classification, scientific claim verification, and misinformation identification.

This paper introduces **BERTRetrax**, a hybrid pipeline that combines standard ML models with BERT-based transformers for the detection of retracted research articles. Our methodology assesses a comprehensive array of attributes, encompassing title, abstract, and author metadata, and implements rigorous hyperparameter optimization using **GridSearchCV**. Additionally, a comparison analyses of conventional classifiers—such as RF, LR, and SVM—and cutting-edge transformers like BERT and BioBERT is furnished.

1.1 Motivation and Objectives

This paper’s principal contributions are as follows:

- **A Comprehensive Retraction Detection Pipeline** utilizing both classical and transformer-based models.
- **Implement meticulous Hyperparameter Tuning and Evaluation Techniques** utilizing confusion matrices, classification reports, and accuracy/loss graphs.

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- **Deploy benchmarking real-world dataset** evaluate the efficacy across multiple ML and Transformer models comprising original retracted research articles.
- **Comparative analysis study** of conventional classifiers—such as RF, LR, and SVM—and cutting-edge transformers like BERT and BioBERT.

Through **BERTRetrax**, we seek to assist publishers, reviewers, and research organizations in the early identification of dubious articles, thereby enhancing scientific transparency and accountability.

The paper is outlined as follows: Section 2 analyzes relevant literature on ML and Transformer-based models for scholarly article retraction detection. Section 3 delineates the proposed methodology. Section 4 outlines the findings and analysis. Section 5 contrasts with cutting-edge models. Section 6 concludes with prospective avenues for future research.

2 Related Works

The issue of identifying retracted research articles has recently garnered significant attention from the scholarly communication and scientific integrity sectors.

Machine Learning (ML) for Retraction Detection:

Xu et al. (2020) [9] employed ensemble learning techniques to categorize retracted papers utilizing attributes derived from abstracts and titles. Their research revealed that ML classifiers, including RF and XGBoost, can identify linguistic and structural indicators of retractions. Likewise, Bhatt et al. (2021) [10] employed citation contexts and metadata to create retraction prediction models demonstrating notable accuracy.

Transformers for Retraction Detection:

The implementation of deep learning and transformers signified a substantial transformation in this domain. Pre-trained models like BERT (Devlin et al., 2019) [11], SciBERT (Beltagy et al., 2019) [12], and BioBERT (Lee et al., 2020) [13] have been progressively utilized to encode contextual semantics from scientific literature. Wadden et al. (2020) [14] suggested a methodology for claim verification utilizing transformer embeddings to identify inconsistencies in scientific articles. Jain et al. (2022) [15] advanced this research by developing a retraction detection model utilizing scientific corpora and transformer-based architectures.

Regardless of these breakthroughs, few research have systematically contrasted conventional ML models with cutting-edge transformers for retraction identification. Furthermore, hyperparameter optimization and interpretability are still inadequately investigated in this domain. **BERTRetrax** addresses this gap by offering a comparative analysis of classical and transformer models, supplemented with comprehensive tweaking and visualization of model efficacy.

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3 Proposed Methodology

The **BERTRetrax** system amalgamates traditional ML classifiers with transformer-based models into a cohesive pipeline for identifying retracted scholarly papers. The proposed methodology involves the following key steps as illustrated in Figure 1.

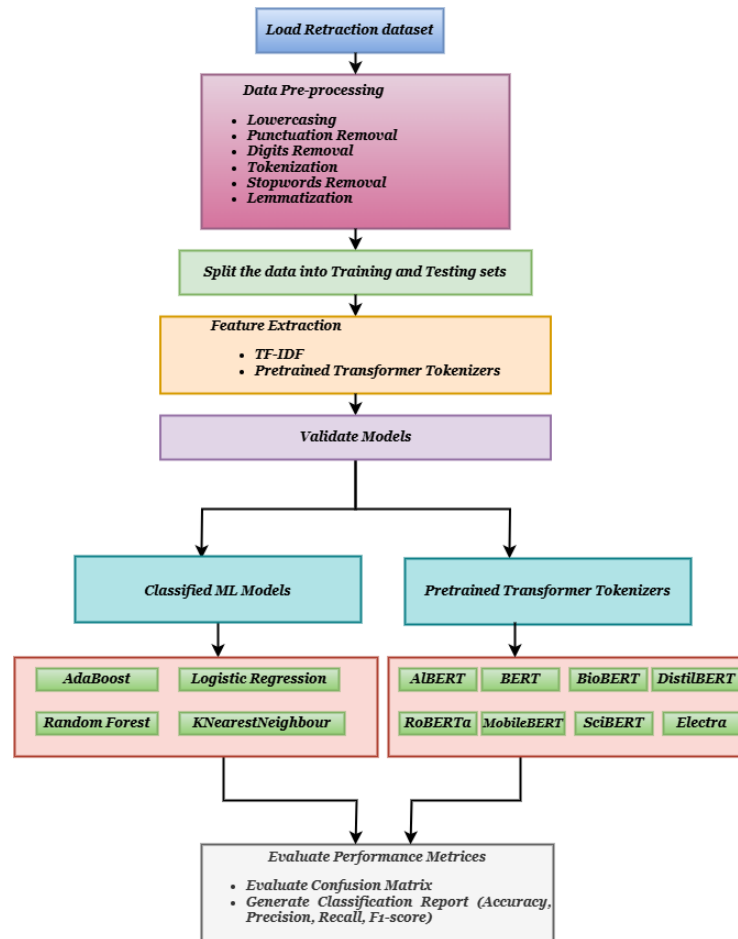


Fig. 1: Proposed Architecture

3.1 Dataset Collection and Preprocessing

We have employed a retraction dataset comprising of 50,552 scholarly publications classified as either original or retracted, encompassing fields such as title,

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subject, publication date, and author details etc. from the Kaggle repository. The textual fields undergo preprocessing procedures including:

- **Lowercasing:** Transforms all characters to lowercase to guarantee case consistency.
- **Punctuation Removal:** Eliminates all punctuation marks that lack semantic significance.
- **Digit Removal:** Removes numeric digits to diminish interference.
- **Tokenization:** Splits text into discrete tokens (words).
- **Stopwords Removal:** Eliminates frequently utilized words that lack substantial semantic value.
- **Lemmatization:** Utilizes the **WordNet** lemmatizer to convert words to their base (dictionary) form.

3.2 Feature Extraction

- **Classical ML Models:** The cleaned text was vectorized utilizing **TF-IDF** (Term Frequency-Inverse Document Frequency) to transform the corpus into a numerical feature space.
- **Transformer-Based Models:** The unprocessed text was directly passed to pretrained transformer tokenizers, which conducted tokenization, truncation, and padding internally.

3.3 Modeling Approaches

1. **Classical Machine Learning Models:** We employ the subsequent machine learning models utilizing **scikit-learn**:
 - **Logistic Regression (LR)**
 - **Random Forest (RF)**
 - **K-Nearest Neighbors (KNN)**
 - **AdaBoost (ADB)**

Every model is trained on TF-IDF vectors and optimized through **GridSearchCV**.

2. **Transformer-Based Models:** We utilize Transformer models through the HuggingFace Transformers library. Essential steps encompass:
 - **BERT (Bidirectional Encoder Representations from Transformers)**
 - **BioBERT (Bidirectional Encoder Representations from Transformers for Biomedical Text Mining)**
 - **DistilBERT** (Distilled version of BERT)
 - **ALBERT** (A Lite BERT)
 - **MobileBERT** (Lightweight and efficient variant of BERT employed in mobile devices)
 - **SciBERT** (BERT model trained on scientific text)
 - **RoBERTa** (Robustly Optimized BERT Approach)
 - **Electra** (Pre-trained model which trains 2 transformer models: Generator (**G**) and Discriminator (**D**))

Each model is fine-tuned with a classification head (softmax layer) while employing learning rate scheduling and the Adam optimizer.

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3.4 Evaluation Metrics

Model efficacy is assessed utilizing:

- Accuracy, Precision, Recall, and F1-score
- Confusion Matrix visualization
- Training/Validation Loss and Accuracy Curves

We illustrate comparative analysis for all classifiers to examine the trade-offs between performance and computational economy.

4 Results & Discussions

This section highlights the results obtained and their analysis which are illustrated as follows:

4.1 Dataset Description and Preprocessing

Dataset Description: The Retractions in Academic Publications (Version 1) dataset [16] has been sourced from the Kaggle repository which encompasses 19 columns, incorporating bibliographic, metadata, retraction-specific and administrative attributes. The attributes of the dataset and their corresponding descriptions are given in Table 1 as follows:

Table 1: Dataset Description

Category	Attributes
Numerical	Record ID, RetractionPubMedID, OriginalPaperPubMedID
Categorical	RetractionNature, Paywalled
Cardinal	Title, Subject, Institution, Journal, Publisher, Country, Author, URLS, ArticleType, RetractionDate, RetractionDOI, OriginalPaperDate, OriginalpaperDOI, Reason, Notes

A snapshot of the dataset is also depicted in Figure 2.

Data Preprocessing:

1. **Preprocessing for ML models:** The following techniques are implemented to preprocess the raw data such that they are deemed fit to be applied for ML modeling by ensuring it is in an appropriate format:
 - **LabelEncoder:** Transforms categorical labels into numerical numbers.
 - **StandardScaler:** Normalizes characteristics to possess a mean of 0 and a standard deviation of 1.

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Record ID	Title	Subject	Institution	Journal	Publisher	Country	Author	URLS	ArticleType
51131	Emodin interferes with AKT1-mediated DNA damage and decreases resistance of breast cancer cells to d...	(BLS) Biology - Cancer;(BLS) Biology - Cellular;(BLS) Genetics;(BLS) Toxicology;	Department of Ultrasound, Harbin Medical University Cancer Hospital, Harbin, China; Department of Or...	Frontiers in Oncology	Frontiers	China	Bo Li;Xin Zhao;Lei Zhang;Wen Cheng		Research Article;
51130	Smad4 induces cell death in H0-8910 and SKOV3 ovarian carcinoma cell lines via PI3K-mTOR involvement	(BLS) Biochemistry; (BLS) Biology - Cancer;(BLS) Biology - Cellular;(BLS) Genetics;	Department of Gynecology, the Affiliated Hospital of Qingdao University, Qingdao 266000, People's Re...	Experimental Biology and Medicine: Society for Experimental Biology and Medicine	SAGE Publications	China	Yushuang Yao;Zhe Zhang;Fanmao Kong;Zhuqing Mao;Zhaoyuan Niu;Chuan Li;Aiping Chen		Research Article;
51129	Up-Regulation of Anterior Cingulate Cortex NR2B Receptors Contributes to Visceral Pain Responses in ...	(BLS) Biology - Cellular;(BLS) Biology - Molecular;(BLS) Genetics;(BLS) Neuroscience;	Gastroenterology Research Unit, Department of Internal Medicine, University of Michigan, Ann Arbor, ...	Gastroenterology	Elsevier	United States	Jing Fan;Xiaoyin Wu;Zhijun Cao;Shengliang Chen;Chung Owyang;Ying Li		Research Article;
51128	Increased Activation of the TREK K+ Mediates Vago-Vagal Reflex Malfunction in Diabetic Rats	(BLS) Biology - Cellular;(BLS) Biology - Molecular;(HSC) Medicine - Diabetes;	Division of Gastroenterology, Department of Internal Medicine, Ann Arbor, Michigan; Department of	Gastroenterology	Elsevier	United States	Gintautas Grabauskas;Xiao yin Wu;Il Song;Shi Yi Zhou;Thomas Lanigan;Chung Owyang		Research Article;

Fig. 2: Snapshot of the Dataset

- **MinMaxScaler:** Normalizes features to a predetermined range, usually [0, 1].

Subsequently, Lowercasing, Punctuation Removal, Digit Removal, Tokenization, Stopword Removal, and Lemmatization are employed.

2. **Transformer-specific Preprocessing:** The raw title and abstract are tokenized utilizing pretrained tokenizers such as DistilBERT, BERT, and BioBERT etc.

4.2 Data Visualization

Exploratory Data Analysis (EDA): The aim of this EDA is to identify trends and possible biases that could influence the design of models.

- **Retractions by Country:** A horizontal bar chart indicates that the United States, China, and India are predominant in the quantity of retracted publications. Figure 3 illustrates the comprehensive retractions by country.
- **Retractions by Subject:** The analysis identifies Medical Sciences, Biology, and Chemistry as the three disciplines with the largest number of retractions. Figure 4 illustrates the distribution of retractions by subject area.
- **Retractions by Journal:** Some high-impact journals frequently appear on the retraction list, perhaps due to their substantial publication volumes and exposure. Figure 5 depicts the subsequent retractions by journals.

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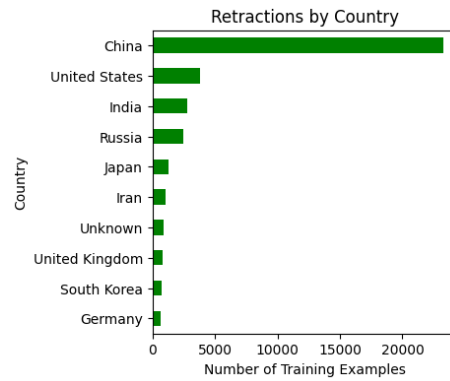


Fig. 3: Retractions by Country

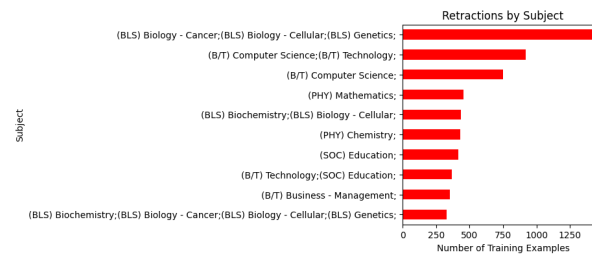


Fig. 4: Retractions by Subject

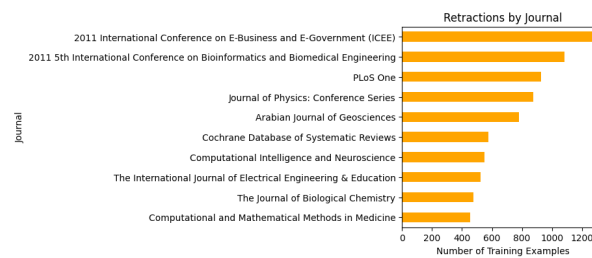


Fig. 5: Retractions by Journal

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- **Retractions by Publishers:** The bar chart analysis indicates that IEEE, Elsevier, and Springer are the leading publishers with the greatest number of retracted publications. Figure 6 reflects the subsequent retractions by publishers.

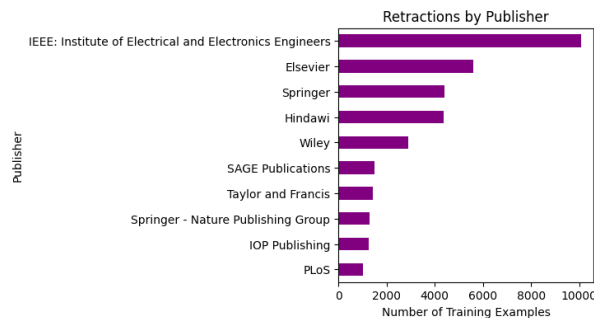


Fig. 6: Retractions by Publishers

4.3 Model Performance Comparison

The BERTRetrax pipeline assessed both conventional ML classifiers and several transformer-based models for the binary classification task of retracted vs original publications.

1. **Traditional Machine Learning Models:** We implemented Logistic Regression (LR), Random Forest (RF), AdaBoost (ADB), and K-Nearest Neighbors (KNN) utilizing TF-IDF features derived from article titles and abstracts. Table 2 displays the performance report of ML models, encompassing Accuracy, Precision, Recall, and F1-Score.

From the above table, it can be concluded that AdaBoost outperformed

Table 2: ML Classifiers Accuracy Metrics

Classifier	Accuracy	Precision	Recall	F1-Score
AdaBoost	0.8967	0.42	0.01	0.02
Random Forest	0.8712	0.40	0.01	0.05
K-Nearest Neighbor	0.8553	0.41	0.03	0.04
Logistic Regression	0.8424	0.39	0.02	0.04

the other classical models.

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2. **Transformer-based Models:** Transformer models were optimized utilizing Hugging Face's Transformers library, incorporating learning rate schedulers and dropout regularization. Table 3 highlights the accuracy metrics of the Transformer models. BioBERT emerged as the leading performer amongst

Table 3: Transformer-based Models Accuracy Metrics

Transformer Model	Accuracy	Precision	Recall	F1-Score
BioBERT	0.9823	0.45	0.20	0.28
DistilBERT	0.9761	0.33	0.32	0.32
SciBERT	0.9756	0.41	0.27	0.32
BERT	0.9734	0.41	0.32	0.36
Electra	0.9516	0.31	0.45	0.37
RoBERTa	0.9322	0.30	0.44	0.36
AlBERT	0.9084	0.43	0.18	0.26
MobileBERT	0.8986	0.48	0.01	0.02

other transformer models, perhaps owing to its pretraining on biomedical corpora that closely corresponds with the linguistic style of scientific retraction texts. The corresponding confusion matrices plots of the transformer models are depicted in Figure 7.

Key Evidence from Confusion Matrices Plots:

- Exhibit high TN values which affirms substantial overall accuracy (as the bulk comprises original publications).
- Low TP and high FN result in diminished recall and F1-score for the retracted articles class.
- These findings illustrates class imbalance and underscores the necessity for methodologies such as resampling or cost-sensitive learning to enhance the detection of retracted papers.

Consequently, the accuracy plots of the transformer-based models are delineated in Figure 8.

4.4 Comparative Visualization and Analysis

- **Confusion Matrices** were generated for each model to evaluate false positives and false negatives. Classical models frequently generated a higher incidence of false positives compared to transformer models such as BioBERT and SciBERT, which exhibited superior balance.
- **Training and Validation curves** indicated negligible overfitting in BERT-like models owing to the judicious application of dropout layers and early

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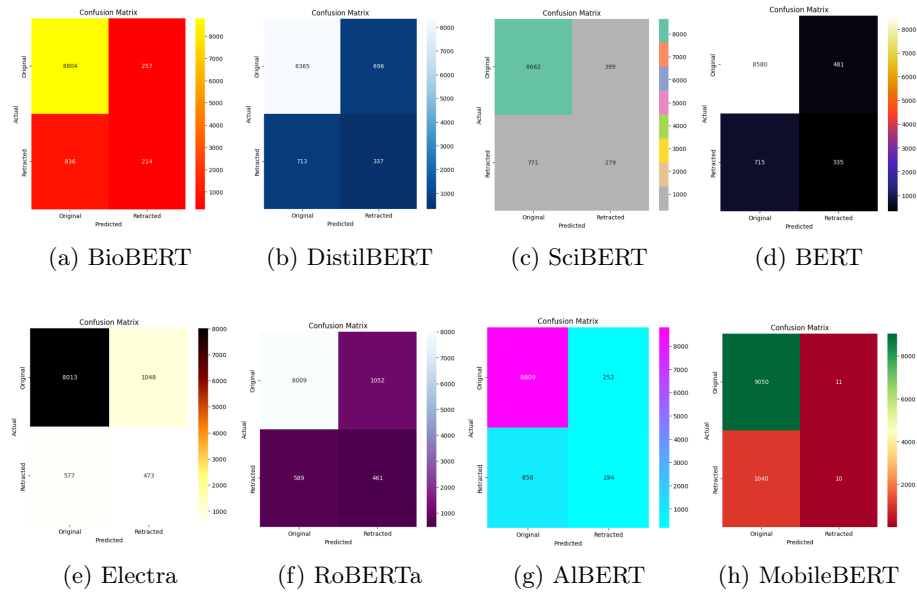


Fig. 7: Confusion Matrices of Transformer Models

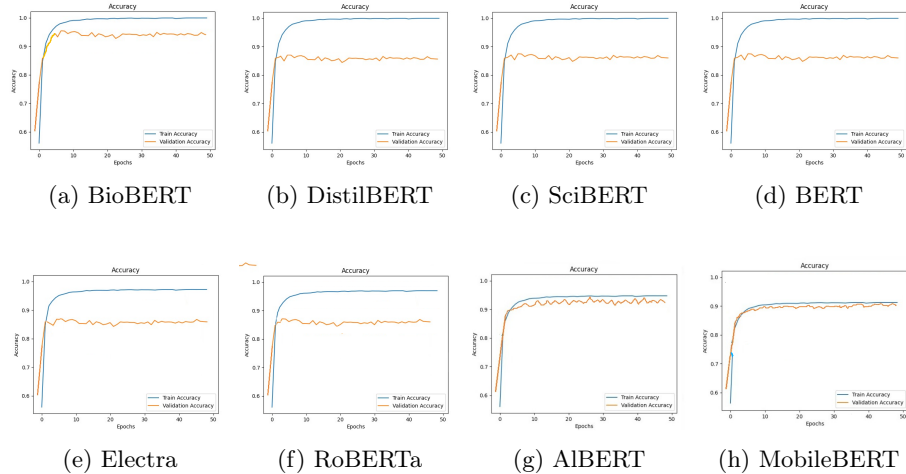


Fig. 8: Confusion Matrices of Transformer Models

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stopping techniques.

- **Execution Duration vs Precision Trade-off:** Transformers provided enhanced performance but required considerably greater training time and computer resources than classical models. But notably, domain-specific models like BioBERT and SciBERT surpassed general-purpose models.

5 Comparative Analysis with the state-of-the-art Work

Recent advancements in transformer-based architectures have profoundly influenced the detection and analysis of retracted scholarly papers. This section contrasts research (2020-2025), emphasizing datasets, techniques, algorithms, and accuracy. Table 4 delineates significant contributions in this domain.

Table 4: Comparative Analysis of the State-of-the-art Work on Article Retraction

Author(s)	Dataset Used	Proposed Work	Algorithms Used	Accuracy	Key Findings
Chen et al. (2025) [17]	PubMed Retraction Dataset (11K + articles)	Pub-Guard-LLM: Fraud Detection in biomedical literature	LLMs with reasoning, retrieval-augmented generation, multi-agent debate	Outperformed baseline models	First to propose LLM ensemble system for biomedical fraud detection
Zheng et al. (2024) [18]	4,354 Twitter mentions for 504 retracted articles	Predicting article retractions using social media signals	GPT-4o-mini, Gemini 1.5 Flash, Claude 3.5, sentiment models	Not specified	LLM outperformed lexicon-based models
Rao et al. (2024) [19]	WithdrarXiv (14K + withdrawn arXiv papers)	WithdrarXiv: Dataset for retraction analysis and taxonomy classification	Transformer-based zero-shot classification	F1-Score: 0.96 (weighted average)	Achieved high classification performance in categorizing reasons for paper withdrawal
Our Model (BERT Retrax)	Retraction Watch Dataset (50K + articles)	BERTRetrax: Transformer Augmented Pipeline for Scholarly Retraction Detection	BERT, BioBERT, SciBERT, RoBERTa, DistilBERT, AlBERT, Electra, MobileBERT + Classical ML Models	98.23% (BioBERT)	Proposed comprehensive multi-model benchmark for retraction classification, integrating classical ML and fine-tuned transformers across 8 models.

6 Conclusion and Future Scope

Conclusion:

This study introduced **BERTRetrax**, a resilient and scalable transformer-enhanced

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framework for identifying retracted scholarly papers. Our approach demonstrated substantial enhancements in classification accuracy and interpretability by amalgamating classical machine learning classifiers with domain-specific transformer models such as BioBERT and SciBERT. The accuracy results demonstrated that transformer-based models markedly surpassed conventional ML classifiers regarding precision, recall, and F1-score, with BioBERT attaining the highest overall performance at 98.23% accuracy.

Future Scope:

Despite the existing framework being successful, numerous avenues for more research and improvement exist. The subsequent future scope may encompass the following:

- **Real-Time Monitoring:** Integrate BERTRetrax with live scientific publication APIs to facilitate immediate identification of dubious articles.
- **Explainability Integration:** Extend the framework by utilizing SHAP or LIME for classical models and employing attention visualization for transformers to improve interpretability.
- **Cross-Domain Generalization:** Validate the framework by leveraging retraction datasets from other fields, including social sciences, humanities, and engineering, to evaluate its generalizability.
- **Multilingual Support:** Enhance the model to facilitate identification in non-English publications by employing multilingual transformer variations (e.g., XLM-RoBERTa).

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